Rebuttal for Submission 230

Dataset	s=1	s=2	s=4	s=8	s=16
UCF-101	0.395	0.395	0.395	0.395	0.617
AUTSL	0.125	0.125	0.131	0.252	0.475

Table 1. Median seconds between selected frames at 100% coverage.

We thank the reviewers. We agree that our original terminology and some mechanistic language were too strong. The requested checks below distinguish fixed-budget sampling sensitivity from literal signal aliasing, while preserving the dataset-ranking result.

TDS and architecture claim (R2). TDS is not an endpoint artefact: replacing the endpoint drop with the full stride-curve integral gives the identical ranking (Spearman 1.000), the positive part never activates, and baseline/headroom normalisation only swaps adjacent datasets (0.952). We also ran the requested matched-evidence control: every model receives the same k distinct positions before resampling to its native input length. At $k = 2$, its accuracy ranking agrees with the original ordering (Spearman 0.857, p=0.007, all 8 models); agreement is weaker at the other budgets. Thus the evidence supports a robust dataset ranking and a narrow observational architecture association, not the prior 3–5× claim or a causal temporal-module explanation.

What stride changes (R2, R3). Every model consumes a fixed budget B. Strided candidates are re-uniformised to B while sufficient candidates remain, then repeat-last padded. The sweep therefore changes distinct evidence as well as nominal stride; we will rename it sampling sensitivity and report the realised intervals in seconds:

For clips that never shortfall, the mean s=1 → 16 change is −0.5pp versus +10.7pp overall (21/21 usable model–dataset pairs); matched evidence reduces the frame-budget/drop correlation from 0.849 (p = .008) to 0.231 (p = .58). Pinning frame count while varying only span changes realised rate tenfold yet raises accuracy in 7/8 datasets, showing that context and evidence must be separated. Measured frame rates also make a fixed nominal stride vary 3.9× in realised interval across datasets.

Training, spatial protocol, and statistics (R1, R2, R3). The requested multi-stride ablation confirms a training–inference mismatch: exposure recovers 28.4% of the TimeSformer drop and 37.0% for R3D-18 (6.5%→31.9% at stride 16), but costs 3.3/9.9pp dense accuracy and leaves 63–72% of the drop. The zero-shot spatial sweep is an input-scale resampling test, not new-grid Transformer transfer; we will state the exact resize/preprocessing protocol and add no-fine-tuning baselines. At 48px, 52.8% of 503 runs peak in the final epoch, but the final-three-epoch gain is only 1.38pp, so we will report this convergence limitation. Repeated-measures ANOVA gives partial η^2=0.292 (coverage), 0.232 (stride), and 0.070 (interaction). Kinetics-400 is the shared pretraining source, not a new evaluation domain; we will clarify this, add FineGym where requested, and make model ordering consistent.

Routing (R2). The rule thresholds $\max_c p(c|v)$, not entropy. With held-out calibration (1000 stratified splits), SSv2/TimeSformer changes from +0.21pp in-sample to −0.03pp (95% CI $[-1.07, +0.83]$): it matches dense inference at 6.9 frames. We therefore rename it a confidence cascade and remove the general "no penalty" mitigation claim; over 56 model–dataset pairs its mean change is −3.5pp. Device-scaling extrapolations are removed.